Programming for Data Analytics

Part 2 — Classification and Model Improvement

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1. Justification of dataset

Dataset: Breast Cancer Wisconsin (Diagnostic), UCI Machine Learning Repository (Wolberg, Street and Mangasarian, 1995). It holds 569 breast masses, each described by 30 numeric measurements from a digitised fine needle aspirate image, and a diagnosis of malignant or benign.

Criterion	Justification
Quality	No missing values and no duplicate rows (checked in the EDA). Labelled by diagnosis, fully documented, widely used and peer-reviewed, so results can be benchmarked.
Suitability	Binary malignant/benign outcome matches the scheme's question of whether a cancer benefit applies. All predictors are numeric, which suits a classification algorithm without complex encoding.
Size	569 rows with 30 features is large enough for a stable train/test split and cross-validation, and small enough to train quickly and explain.
Ethics	Public, anonymised data with no personal identifiers.
Limitation	It covers breast cancer only and comes from one source. The scheme should validate any model on its own claims data before real use.

2. Justification of algorithm

Chosen algorithm: Logistic regression, with standardised inputs.

- **Interpretable.** Each coefficient shows how a feature changes the odds of malignancy. A medical scheme must be able to explain and defend a decision that affects a customer's benefit.
- **Fits the brief.** It outputs a probability, so the scheme can set a decision threshold that reflects the cost of a missed cancer.
- **Supports p-value feature selection.** The brief suggests backward/forward elimination, which needs a model with coefficient p-values.
- **Strong on this data.** The classes are close to linearly separable, so a simple model performs well and complex models (e.g. random forest, neural network) would be harder to explain and are unlikely to add much when a simple model already performs well.

Trade-off: logistic regression assumes a linear relationship on the log-odds scale and is affected by correlated features, so correlation was handled during feature selection and regularisation was tuned.

3. Analysis plan

Stage	Plan
a. EDA	Check shape, types, missing values, duplicates. Class counts. Summary statistics, skewness, IQR outlier counts. Distributions and boxplots by class. Correlation with target and between features.
b. Feature selection	Split first (stratified 80/20) so the test set is never used for selection. On training data: drop one feature of each pair with |r| above 0.8, then backward elimination on logistic-regression p-values (drop highest p until all are below 0.05).
c. Train model	Standardise inside a pipeline. Baseline: default parameters, all features. Then tune by 5-fold stratified cross-validation (grid search): C = 0.01, 0.1, 1, 10, 100; penalty = L1 or L2; class weight = none or balanced.
d. Evaluate	Primary: recall for malignant (missed cancers are the costly error) and F2. Supporting: precision, F1, ROC-AUC, confusion matrix, cross-validation, and a single final test-set score.
e. Report	Data and cleaning, EDA visuals, features chosen, how models were trained, how results were interpreted, what was changed to improve, limitations and recommendation.

4. Report to the client

4.1 Purpose and data

The medical scheme wants to speed up the assessment of dreaded-disease (cancer) benefit applications. This report describes a model that classifies a tumour as malignant or benign from 30 measurements, how it was built, and how reliable it is. It is a decision-support tool and does not replace clinical assessment.

The data contain 569 tumours: 357 benign and 212 malignant. Malignant is treated as the positive class because detecting cancer is the objective.

4.2 Data cleaning and exploratory analysis

- Data quality: no missing values and no duplicate rows, so no imputation or row removal was needed. All 30 predictors are numeric.
- Outliers: 171 of 569 rows have at least one value outside 1.5 times the interquartile range. These were kept because they are real, often large malignant masses, and carry diagnostic signal.
- Skew and scale: 22 of 30 features are noticeably skewed (absolute skewness above 1) and features sit on very different scales (e.g. area in the hundreds, smoothness near 0.1). Features were therefore standardised before modelling.
- Class balance: benign is the majority (357 against 212). This was handled by stratified splitting and by testing a balanced class weight.

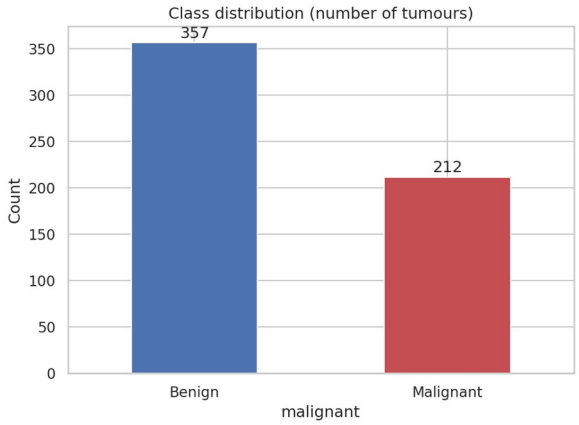


Figure 1: Number of tumours by class

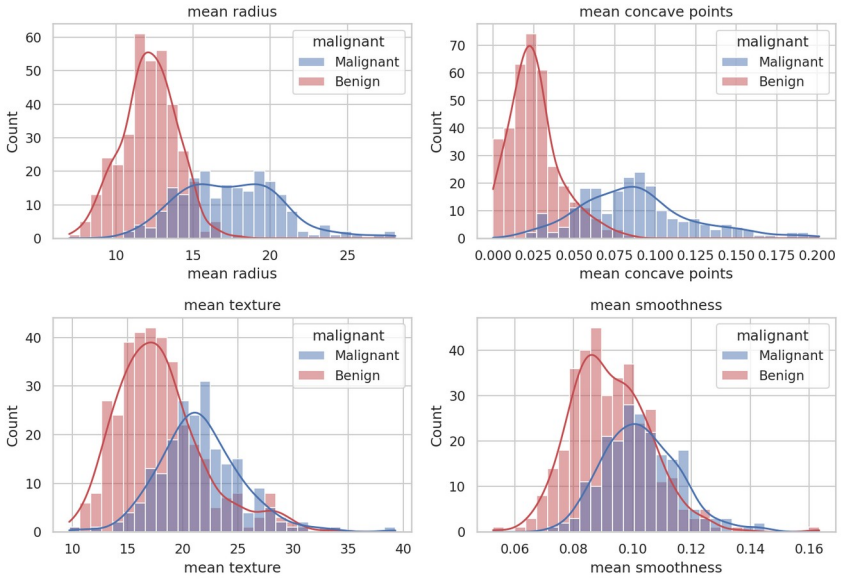


Figure 2: Malignant tumours have clearly higher values of size and concavity measures, with limited overlap

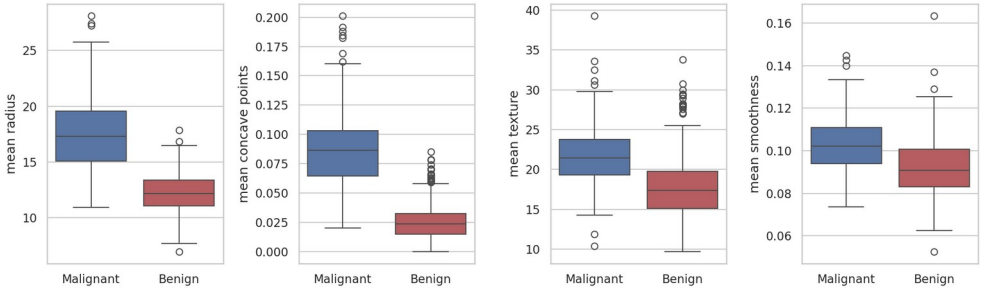


Figure 3: Spread of four features by class

Relationship with the outcome: the strongest correlations with malignancy are worst concave points (0.794), worst perimeter (0.783) and mean concave points (0.777). Many features measure the same thing in different ways: 21 feature pairs have a correlation above 0.9 (e.g. radius, perimeter and area), which is multicollinearity.

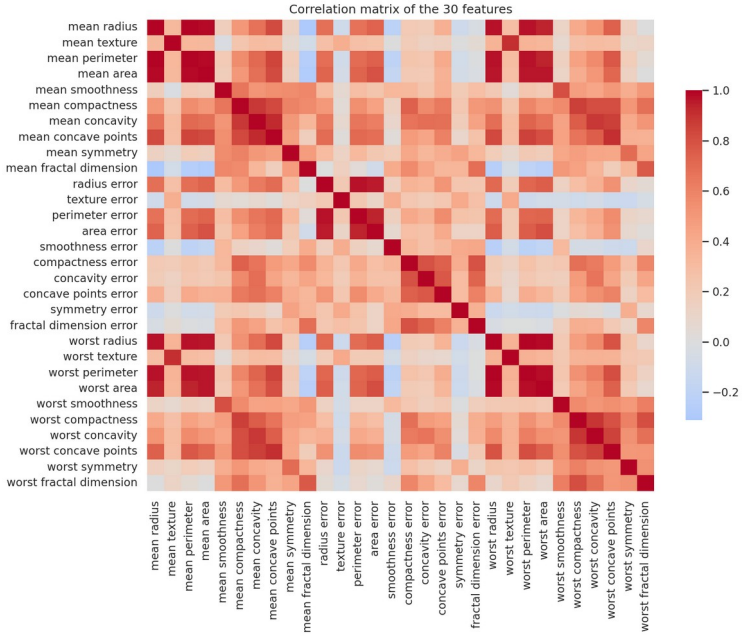


Figure 4: Correlation between features (red = strongly positive)

4.3 Feature selection

Data were split into 455 training and 114 test tumours (stratified; test set held back until the end). On the training data only, 18 features were removed as near-duplicates (correlation above 0.8, keeping the one more related to the outcome), leaving 12. A cut-off of 0.9 was tried first but left 20 correlated features, the logistic model became near-perfectly separable and did not converge, so its p-values could not be trusted.

Backward elimination then removed 6 features in turn: concave points error (p = 0.9421), worst smoothness (p = 0.6461), symmetry error (p = 0.1993), mean fractal dimension (p = 0.0856), mean symmetry (p = 0.0521), worst symmetry (p = 0.4196). Six features remained, all with p below 0.05: radius error, texture error, smoothness error, compactness error, worst texture, worst concave points.

Important finding: this 6-feature set turned out weaker than using all features with regularisation (section 4.5). The correlation filter removed strong size features such as worst radius and worst area, so the p-value route kept some weaker predictors.

4.4 How the models were trained

Model	Features and parameters	Purpose
Model 1	All 30 features, default logistic regression (C = 1, L2)	Baseline
Model 2	6 selected features. Tuned: C = 1, L1, class weight balanced	Test whether p-value selection helps
Model 3	All 30 features. Tuned: C = 0.1, L2, class weight balanced	Test whether tuning and regularisation help
Model 4	Model 3 with the decision threshold lowered from 0.5 to 0.35	Reduce missed cancers

Standardisation was done inside a pipeline so that, during cross-validation, the scaler learned only from the training folds (this prevents data leakage). Tuning used 5-fold stratified cross-validation, scored on F2.

4.5 How the models were evaluated

A missed cancer (false negative) is much more costly than a false alarm (false positive), which can be reviewed by a clinician. Therefore recall for the malignant class and F2 (which weights recall twice as much as precision) are the main metrics. Accuracy alone can hide missed cancers when classes are unequal, so precision, F1, ROC-AUC and the confusion matrix are reported too.

Cross-validation on the training data (mean of 5 folds) was used to choose between models:

Metric	Model 1	Model 2	Model 3
Accuracy	0.9736	0.9626	0.9758
Precision	0.9771	0.9428	0.9766
Recall (sensitivity)	0.9529	0.9588	0.9588
F1	0.9640	0.9503	0.9674
F2	0.9572	0.9553	0.9622
ROC-AUC	0.9958	0.9920	0.9949

Table 1: 5-fold cross-validation results on training data (Models 1 to 3)

Model 3 was best on cross-validated F2 (0.9622), against 0.9572 for the baseline and 0.9553 for Model 2. The differences are small, because the baseline is already strong.

Final scores on the untouched test set (114 tumours: 72 benign, 42 malignant):

Metric	Model 1	Model 2	Model 3	Model 4
Accuracy	0.9649	0.9474	0.9912	0.9561
Precision	0.9750	0.9500	1.0000	0.9111
Recall (sensitivity)	0.9286	0.9048	0.9762	0.9762
F1	0.9512	0.9268	0.9880	0.9425
F2	0.9375	0.9135	0.9809	0.9624
ROC-AUC	0.9960	0.9907	0.9977	0.9977

Table 2: Test-set results (decision threshold 0.5 for Models 1 to 3; Model 4 uses the lower threshold)

Model	True negatives	False positives	False negatives	True positives
Model 1	71	1	3	39
Model 2	70	2	4	38
Model 3	72	0	1	41
Model 4	68	4	1	41

Table 3: Test-set confusion matrix counts

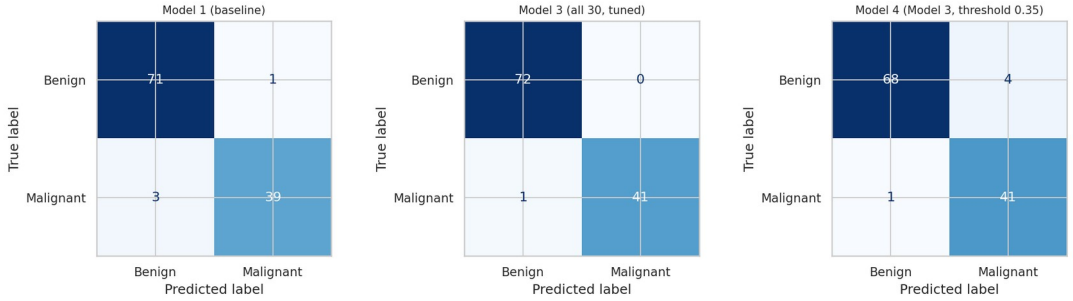


Figure 5: Confusion matrices on the test set

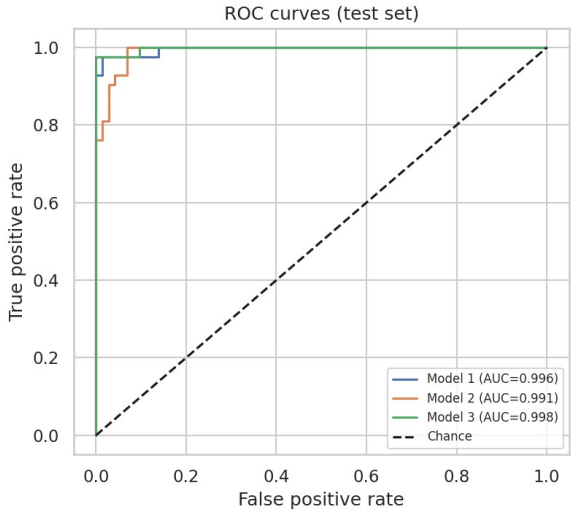


Figure 6: ROC curves on the test set (area under the curve near 1 means near-perfect ranking)

4.6 Retraining and improvement

- Shortcoming of the baseline: it missed 3 of 42 malignant tumours on the test set (and 8 of 170 in cross-validation on the training data), with 1 false alarm.
- Change 1 (Model 3): tuning regularisation and using balanced class weights lifted test recall from 0.9286 to 0.9762, cutting missed cancers from 3 to 1 with no false alarms.
- Change 2 (Model 4): lowering the threshold to 0.35 was chosen from training-data cross-validation, where it cut missed cancers from 7 to 3 (out of 170) at the cost of more false alarms (4 to 11). On the test set it did not help further: missed cancers stayed at 1 while false alarms rose to 4.
- Model 2 (selected features) did not improve on the baseline. This was kept in the report as a genuine result: feature selection by p-value is not automatically better than regularisation.

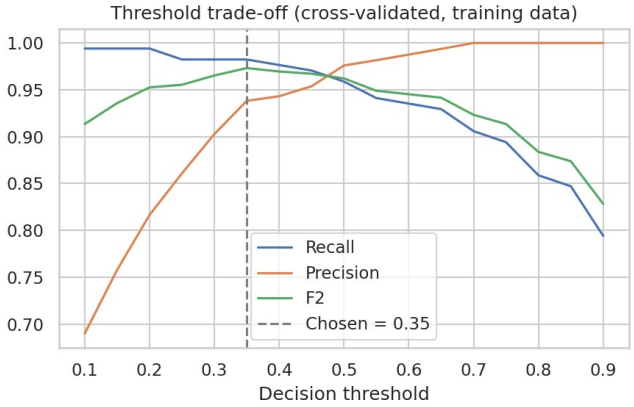


Figure 7: Effect of the decision threshold (cross-validated, training data)

4.7 Interpretation, limitations and recommendation

Model 3 is the recommended model. It had the best cross-validated F2 and the best test results (accuracy 0.9912, recall 0.9762, precision 1.0000, ROC-AUC 0.9977). The strongest drivers of a malignant prediction are worst texture, worst radius, radius error and several concavity and perimeter measures (Figure 8).

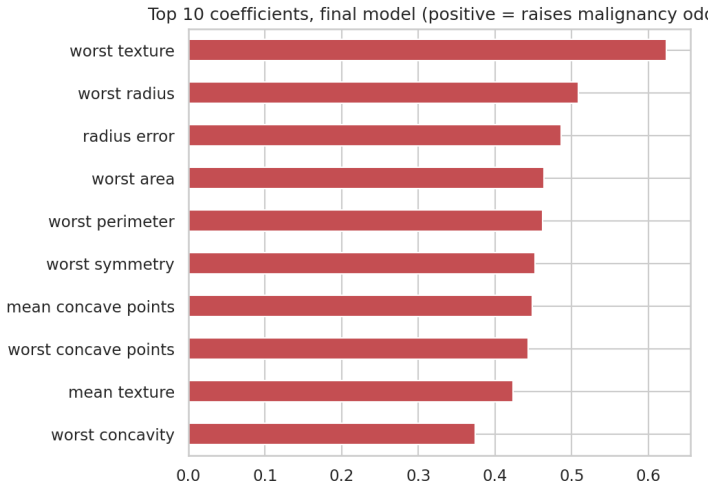


Figure 8: Largest coefficients in Model 3 (standardised features)

- **Expect slightly lower real-world performance.** The test set is small, so one error moves accuracy by about 0.009. Cross-validated accuracy (0.9758) is a more stable estimate than the test score.

- **Threshold is a business decision.** Model 3 and Model 4 performed the same on missed cancers in the test set. If the scheme wants to minimise missed cancers above all, the lower threshold is the safer policy in cross-validation, but it increases the number of cases needing manual review.
- **Scope.** The data cover breast cancer from a single source. The model should be validated on the scheme's own data and other cancer types before use, and any decision should be confirmed by a clinician.

References

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